Week 5: CPU Organization & ALU, Control Unit & Instruction Cycle

Midterm Exam Preparation
Week 5

According to your syllabus, Week 5 has two major parts:

1. CPU Structure and ALU: CPU components, registers, datapath, relationship between CPU components, arithmetic and logical processing.
2. Control Unit and Instruction Cycle: Control Unit, control signals, Instruction Register, Program Counter, fetch-decode-execute cycle, and instruction sequencing.

Let's cover every topic in detail, including examples and important exam questions.

Part 1: CPU Structure and ALU

1. What is a CPU?

CPU (Central Processing Unit) is the main processing unit of a computer. It executes instructions, performs calculations, processes data, and controls the operations of other computer components.

It is often called the brain of the computer because it manages and performs most processing operations.

Main functions of CPU

1. Fetches instructions from memory.
2. Decodes instructions to understand what operation is required.
3. Executes instructions.
4. Performs arithmetic and logical operations.
5. Controls the flow of data between different components.

Example:

Suppose you write:

```
A = 10
B = 20
C = A + B
```

The CPU processes the addition operation and stores the result:

```
C = 30
```

2. Main Components of CPU

The CPU mainly consists of three important components:

1. Arithmetic Logic Unit (ALU)
2. Control Unit (CU)
3. Registers

Notes for A1.1.5 Fetch, Decode, and Execute Cycle - IB | RevisionDojo
revisiondojo.com

A. Arithmetic Logic Unit (ALU)

Definition: The Arithmetic Logic Unit (ALU) is a component of the CPU that performs arithmetic and logical operations on data.

It is responsible for calculations and comparisons.

Arithmetic operations:

- Addition (+)
- Subtraction (−)
- Multiplication (×)
- Division (÷)
- Increment
- Decrement

Logical operations:

- AND
- OR
- NOT
- XOR
- Comparison operations (equal, greater than, less than)

Example 1: Arithmetic operation

```
A = 10
B = 5

ALU performs:
A + B = 15
```

Example 2: Logical operation

```
A = 1010
B = 1100

A AND B = 1000
```

The AND operation produces 1 only when both corresponding bits are 1.

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

Important: ALU performs operations, but the Control Unit tells it which operation to perform.

B. Control Unit (CU)

Definition: The Control Unit is a component of the CPU that directs and coordinates all computer operations by generating control signals.

It manages communication between the ALU, registers, memory, and other components.

Main functions:

1. Fetches instructions from memory.
2. Decodes instructions.
3. Generates control signals.
4. Controls the execution of instructions.
5. Coordinates data movement between CPU components.

Example:

Suppose the instruction is:

```
ADD A, B
```

The Control Unit:

1. Understands that an addition operation is required.
2. Activates the required registers.
3. Sends a control signal to the ALU.
4. ALU performs addition.
5. The result is stored in the destination register.

Remember:

- ALU = Performs operations.
- CU = Controls operations.

C. Registers

Definition: Registers are small, high-speed storage locations inside the CPU that temporarily hold data, instructions, addresses, and intermediate results.

Registers are faster than main memory (RAM).

Why do we need registers?

The CPU frequently needs data while executing instructions. Accessing RAM repeatedly is slower, so registers temporarily store the required information.

Examples:

Suppose:

```
A = 10
B = 20
```

Before addition, the CPU may load these values into registers:

```
Register R1 = 10
Register R2 = 20
```

The ALU performs:

```
R1 + R2 = 30
```

The result can be stored in another register.

Types of registers (general CPU concepts):

Register	Purpose
Data Register	Holds data temporarily
Address Register	Holds a memory address
Instruction Register (IR)	Holds the current instruction
Program Counter (PC)	Holds the address of the next instruction
Accumulator (ACC)	Stores intermediate arithmetic and logical results

Note: Specific CPUs have different register names and organizations. You will study 8086 registers in Week 7.

3. What is Datapath?

Definition: A datapath is the collection of hardware components inside a CPU that moves and processes data.

It includes:

- Registers
- ALU
- Internal buses
- Multiplexers
- Other data-processing circuits

The datapath provides the route through which data travels during instruction execution.

How does a datapath work?

Consider this instruction:

```
ADD R1, R2
```

Assume:

```
R1 = 5
R2 = 3
```

The following operations take place:

1. Data is read from register R1.
2. Data is read from register R2.
3. Both values are sent to the ALU through the datapath.
4. ALU performs addition.
5. The result (8) is sent back to a register.

CS 240 Lab 5
cs.wellesley.edu

Important: The datapath handles the movement and processing of data, while the Control Unit directs the datapath.

4. Relationship Between CPU Components

The three main CPU components work together to execute instructions.

Component	Main responsibility

ALU	Performs arithmetic and logical operations
Control Unit	Controls and coordinates CPU operations
Registers	Temporarily store data and instructions
Datapath	Moves data between CPU components and processes it

Example: How components work together

Suppose the CPU has to execute:

```
R3 = R1 + R2
```

Assume:

```
R1 = 10
R2 = 20
```

Step-by-step process:

1. The Control Unit decodes the instruction.
2. The Control Unit activates the required registers.
3. R1 and R2 provide their values.
4. The datapath transfers both values to the ALU.
5. The ALU performs addition: 10 + 20 = 30.
6. The result travels through the datapath.
7. The result is stored in R3.

Part 4: Assembly Language & Machine Code | Computer Architecture & OS Mastery - Wasil Zafar
wasilzafar.com

Exam point: The CPU components do not work independently. They cooperate to fetch, process, and store information.

5. Arithmetic and Logical Processing

The ALU performs two major categories of operations.

A. Arithmetic Processing

Arithmetic processing involves mathematical calculations.

Examples:

Operation	Example	Result
Addition	10 + 5	15

Subtraction	10 − 5	5
Multiplication	4 × 3	12
Division	20 ÷ 4	5
Increment	5 + 1	6
Decrement	5 − 1	4

Binary addition example:

```
  0101   (5)
+ 0011   (3)
-------
  1000   (8)
```

B. Logical Processing

Logical processing involves operations performed on individual bits.

The main logical operations are:

1. AND

Returns 1 only if both inputs are 1.

```
  1010
AND 1100
--------
  1000
```

2. OR

Returns 1 if at least one input is 1.

```
  1010
OR  1100
--------
  1110
```

3. NOT

Reverses each bit.

```
Input:  1010
Output: 0101
```

4. XOR

Returns 1 when the two corresponding bits are different.

```
  1010
XOR 1100
--------
  0110
```

Truth table for logical operations

A	B	AND	OR	XOR
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

For NOT:

A	NOT A
0	1
1	0

Remember: Arithmetic operations perform calculations, while logical operations manipulate or compare binary data.

Part 2: Control Unit and Instruction Cycle

6. What is the Control Unit?

Definition: The Control Unit (CU) is the part of the CPU that manages and coordinates the execution of instructions by generating control signals.

It does not normally perform calculations itself. Instead, it tells other CPU components what to do and when to do it.

Functions of Control Unit

1. Fetches instructions from memory.
2. Decodes the instructions.
3. Generates control signals.
4. Controls data movement.
5. Coordinates ALU and registers.
6. Maintains the correct sequence of instruction execution.

Example

Consider:

```
ADD R1, R2
```

The Control Unit identifies that an addition is required and sends the necessary control signals to the registers and ALU.

The ALU then performs the addition.

7. What are Control Signals?

Definition: Control signals are electrical signals generated by the Control Unit to direct and coordinate operations inside the CPU.

They tell different components when and how to perform an operation.

Examples of Control Signals

Control Signal	Purpose
Memory Read	Reads data from memory
Memory Write	Writes data into memory
Register Load	Loads data into a register
ALU Control	Selects an ALU operation
PC Increment	Updates the Program Counter
IR Load	Loads an instruction into the Instruction Register

Example: Addition instruction

Suppose:

```
ADD R1, R2
```

The Control Unit generates signals to:

1. Read R1.
2. Read R2.
3. Select addition in the ALU.
4. Enable the destination register.
5. Store the result.

Important: Without control signals, different CPU components would not know when to perform their required operations.

8. Instruction Register (IR)

Definition: The Instruction Register (IR) is a special CPU register that temporarily stores the instruction currently being decoded or executed.

When an instruction is fetched from memory, it is loaded into the IR.

Example:

Suppose memory contains:

```
Address 100: ADD R1, R2
```

When the CPU fetches this instruction:

```
IR = ADD R1, R2
```

The Control Unit decodes the instruction stored in IR and determines what operations are required.

Remember: IR holds the current instruction.

9. Program Counter (PC)

Definition: The Program Counter (PC) is a special CPU register that holds the memory address of the next instruction to be fetched.

It helps the CPU execute instructions in the correct sequence.

Example

Suppose instructions are stored at these memory addresses:

Address	Instruction
100	LOAD R1, 5
101	LOAD R2, 10
102	ADD R1, R2
103	STORE R3

Initially:

```
PC = 100
```

The CPU fetches the instruction at address 100.

After fetching, the PC is updated to point to the next instruction (assuming each instruction occupies one address unit):

```
PC = 101
```

Then it fetches the instruction at address 101, and so on.

Important: The PC stores an address, not the actual instruction.

10. Fetch-Decode-Execute Cycle

Very Important for Exams

Definition: The instruction cycle is the complete process through which the CPU fetches, decodes, and executes an instruction.

Every instruction generally passes through three main stages:

1. Fetch
2. Decode
3. Execute

[Paper 1 Lesson 2.3 - CPU Architecture | Graded IB Computer Science Notes](graded-cs-resources.github.io)
graded-cs-resources.github.io

Stage 1: Fetch

Definition: Fetch is the process of retrieving an instruction from main memory.

Steps:

1. The PC contains the address of the next instruction.
2. The CPU sends this address to memory.
3. Memory returns the instruction.
4. The instruction is loaded into the Instruction Register (IR).
5. The PC is updated to point to the next instruction in a simple sequential execution.

Example:

```
PC = 100

Memory[100] = ADD R1, R2

IR = ADD R1, R2
```

Stage 2: Decode

Definition: Decode is the process of interpreting the fetched instruction to determine what operation needs to be performed.

The Control Unit examines the instruction in the IR.

For example:

```
IR = ADD R1, R2
```

The Control Unit identifies:

- Operation: ADD
- First operand: R1
- Second operand: R2
- Destination: Depends on the instruction format and CPU architecture.

It then generates the required control signals.

Stage 3: Execute

Definition: Execute is the stage in which the CPU performs the operation specified by the instruction.

For example:

```
R1 = 10
R2 = 20

ADD R1, R2
```

The ALU performs:

```
10 + 20 = 30
```

The result is stored in the appropriate destination register.

Complete Example

Suppose memory contains:

```
Address 100: LOAD R1, 10
Address 101: LOAD R2, 20
Address 102: ADD R1, R2
```

For the third instruction:

Fetch:

```
PC = 102
IR = ADD R1, R2
```

Decode:

```
Operation = Addition
Operands = R1 and R2
```

Execute:

```
R1 = 10
R2 = 20

ALU performs 10 + 20 = 30
```

Result:

```
R1 = 30
```

This is a simplified example. Exact register updates depend on the CPU architecture and instruction format.

11. Instruction Sequencing

Definition: Instruction sequencing is the process of determining the order in which instructions are executed by the CPU.

The Program Counter (PC) plays an important role in instruction sequencing.

There are two common types:

A. Sequential Instruction Sequencing

In sequential execution, instructions are executed one after another in their normal order.

Example:

```
Instruction 1
Instruction 2
Instruction 3
Instruction 4
```

The PC normally advances to the next instruction after fetching the current one.

```
PC = 100 → 101 → 102 → 103
```

B. Branching Instruction Sequencing

In branching, the CPU changes the normal sequence of execution and jumps to another instruction address.

This happens when a branch or jump instruction is executed.

Example:

```
Address 100: MOV R1, 5
Address 101: JMP 200
Address 102: ADD R1, 1
Address 200: SUB R1, 1
```

Execution:

1. CPU executes instruction at address 100.
2. It fetches the jump instruction at address 101.
3. The jump instruction changes the PC to 200.
4. CPU executes the instruction at address 200.
5. Address 102 is skipped.

[How much do you know about CPU? - Utmel](#)
[utmel.com](#)

Difference:

Sequential Execution	Branching
Executes instructions in normal order	Changes the normal execution order
PC advances normally	PC is updated to a branch target
No jump is required	Uses jump or branch instructions
Example: 100 → 101 → 102	Example: 100 → 200 → 201

Important Differences for Exams

1. ALU vs Control Unit

ALU	Control Unit
Performs arithmetic and logical operations	Controls CPU operations
Performs calculations	Generates control signals
Processes data	Coordinates components
Example: Addition	Example: Telling ALU to add

2. IR vs PC

Instruction Register (IR)	Program Counter (PC)
Stores the current instruction	Stores the address of the next instruction

Holds instruction data	Holds an instruction address
Used during decoding and execution	Used during instruction fetching

3. Register vs Main Memory

Register	Main Memory (RAM)
Located inside CPU	Located outside CPU
Very fast	Comparatively slower
Very small storage capacity	Larger storage capacity
Stores temporary processing data	Stores currently running programs and data

4. Datapath vs Control Unit

Datapath	Control Unit
Moves and processes data	Directs operations
Includes registers and ALU	Generates control signals
Performs the data-processing work	Coordinates the work

Important Long Questions for Midterm

Prepare these questions carefully:

1. Explain the structure of CPU with a neat, labeled diagram.
2. Explain the functions of ALU, Control Unit, and Registers.
3. What is a datapath? Explain its role in CPU organization.
4. Explain the relationship between CPU components during instruction execution.
5. Differentiate between arithmetic and logical processing with examples.
6. What is a Control Unit? Explain its functions.
7. What are control signals? Explain their role in CPU operations.
8. Explain the functions of Instruction Register (IR) and Program Counter (PC).
9. Explain the Fetch-Decode-Execute cycle with a diagram and example.
10. Explain sequential and branching instruction sequencing with examples.

Short Questions and Answers

Q1. What is CPU?

CPU is the main processing unit of a computer that executes instructions and controls computer operations.

Q2. What is ALU?

ALU is the CPU component that performs arithmetic and logical operations.

Q3. What is CU?

Control Unit coordinates CPU operations and generates control signals.

Q4. What are registers?

Registers are small, high-speed storage locations inside the CPU.

Q5. What is a datapath?

A datapath is the collection of CPU components responsible for moving and processing data.

Q6. What is IR?

IR stores the instruction currently being processed.

Q7. What is PC?

PC stores the address of the next instruction to be fetched.

Q8. What are control signals?

Control signals are electrical signals that direct CPU components to perform specific operations.

Q9. What are the three main stages of the instruction cycle?

Fetch, Decode, and Execute.

Q10. What is branching?

Branching is changing the normal sequence of instruction execution by jumping to another instruction address.

Practice MCQs

Week 5 self-test

0/10 answered

1. Which CPU component performs arithmetic and logical operations?

Control Unit
ALU

Program Counter
Instruction Register

2. Which component generates control signals?

ALU
RAM
Control Unit
Datapath

3. Which register stores the current instruction?

PC
IR
MAR
Accumulator

4. What does the Program Counter store?

Current data
Result of addition
Address of the next instruction
Current ALU operation

5. Which stage interprets an instruction?

Fetch
Decode
Execute
Store

6. Which of the following is a logical operation?

Addition
Multiplication
AND
Division

7. What is the correct basic instruction cycle?

Execute → Fetch → Decode
Decode → Execute → Fetch
Fetch → Decode → Execute
Fetch → Execute → Decode

8. What happens in branching?

Instructions always execute sequentially
The PC changes to a branch target
The ALU stops working permanently
The IR stores RAM capacity

9. Which component moves and processes data?

Datapath
Control Unit only
Hard disk
Keyboard

10. Which CPU component temporarily stores data at high speed?

Registers
Monitor
Hard disk
Printer
Submit answers

Final Revision: What to Memorize

Before your exam, make sure you can explain these points without looking at the notes:

- CPU: ALU + Control Unit + Registers.
- ALU: Arithmetic and logical operations.
- CU: Controls operations and generates control signals.
- Registers: Fast temporary storage inside CPU.
- Datapath: Moves and processes data.
- IR: Stores the current instruction.
- PC: Stores the address of the next instruction.
- Instruction Cycle: Fetch → Decode → Execute.
- Sequential execution: Instructions execute in normal order.
- Branching: Changes the normal execution sequence.

Exam tip: Give special attention to the CPU block diagram, the Fetch-Decode-Execute cycle, and the difference between IR and PC. Practice explaining each diagram step by step, because these topics are suitable for both short and long questions.